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Pathology, virulence-associated gene profiling, antimicrobial susceptibility, and pathogenicity of untypeable capsular serotypes of *Pasteurella multocida* isolated from slaughtered pigs of India

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Abstract

Pasteurella multocida is widely distributed in all pig-rearing countries, affecting the economic viability and profitability of pig production. The present research highlights the molecular characterization and pathology of untypeable capsular serotypes of *P. multocida* in slaughtered pigs from prominent pig-rearing states of India. The prevalence of Pasteurellosis was 27.17% by *Pasteurella multocida* specific Pasteurella multocida specific PCR (PM-PCR). assay, while isolation rate was 7.62%. The microscopic lesions of bronchopneumonia, tonsillitis, and the presence of bacterial antigens in immunohistochemistry confirmed *P. multocida* with pathologies. In capsular typing, the majority of the isolates were untypeable with prevalence of 52.15% and 43.58% in molecular and microbiological methods, respectively. All the isolates showed the uniform distribution of virulence genes such as exxB, nanB, sodC, plpB, and oma87 (100%), while the variations were observed in ptfA, hasR, ptfA, pfhA, hsf-1, and plpE genes. The untypeable isolates showed higher prevalence of hsf-1 gene as compared to others. The untypeable serotypes showed a higher degree of resistance to ampicillin, amoxicillin, and penicillin antibiotics. The mouse pathogenicity testing of untypeable capsular isolates confirmed its pathogenic potential. The higher frequency of pathogenic untypeable isolates with antibiotic resistance profile might pose a serious threat to the pigs, and therefore, preventive measures should be adopted for effective control.

Impact Statement

This study provides valuable information on the prevalence, pathology, serotyping, virulence-associated genes, antimicrobial resistance (AMR), and pathogenicity of prevalent untypeable capsular serotypes of *Pasteurella multocida* isolates from slaughtered pigs, which will guide the clinicians for its effective control in the field.

Keywords: antibiotic sensitivity, capsular typing, *Pasteurella multocida*, pathology, slaughtered pigs, virulence genes

Introduction

Pasteurella multocida, a Gram-negative, nonmotile, and facultative coccobacilli of the Pasteurellaceae family, is responsible for significant economic losses to livestock and wildlife especially in tropical countries (Smith et al. 2021, Truswell et al. 2023). Porcine Pasteurellosis is now recognized worldwide, involving several countries including India (Rajkhowa et al. 2012, Al-Maary et al. 2017, Smith et al. 2021). The pathogen is frequently isolated from the pneumonic lungs of pigs and is thought to play a key role in pneumonic Pasteurel-

losis (Liu et al. 2017, Paladino et al. 2017). Asymptomatic carrier pigs act as a significant source of infection in-contact and susceptible populations (Townsend et al. 2000, Jamaludin et al. 2005). The strains of *P. multocida* are grouped into five capsular serogroups (A, B, D, E, and F), out of which, serotypes A, D, and B are frequently reported to cause diseases in pigs (Liu et al. 2017, Kim et al. 2019). Recently, serotype F has been reported to cause mortality in pigs (Peng et al. 2017, Kim et al. 2019). The majority of the strains isolated from pneumonic lungs belonged to nontoxigenic strains of

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capsular type A (Radanovic et al. 2016, Al-Maary et al. 2017, Liu et al. 2017) and a few from toxigenic strains of capsular type D (Rajkhowa et al. 2012). The changing pattern of the prevalence of various capsular serotypes of *P. multocida* with the disease over time within particular region has been documented in earlier reports (Rajkhowa 2015, Garcia-Alvarez et al. 2017). The various virulence factors play a prime role for the survival of *P. multocida* in the host environment (Rajkhowa et al. 2015, Aski and Tabatabaei 2016, García-Alvarez et al. 2017). The virulence factors such as outer membrane and porin proteins (oma87, ompH, and plpB), adhesins (pfhA, ptfA, hsf-1, and hsf-2), superoxide dismutases (sodA and sodC), iron-acquisition-related factors (exbB, tbpA, hgbA, and hgbB), neuraminidases (nanB and nanH), hyaluronidase, and toxin (toxA) predict the pathogenicity along with the selection of potential future vaccine candidates (Rajkhowa et al. 2015, Sarangi et al. 2015).

Swine Pasteurellosis is endemic in India, causing a considerable degree of mortality in pigs (Rajkhowa et al. 2012, Rajkhowa 2015). Since serotypes vary according to the geographical region and with time in a given region (Liu et al. 2017), there is a need for continuous monitoring of *P. multocida* to record the magnitude of the problem in different areas of the county for its effective control. The present study was carried out to provide baseline data regarding the distribution of capsular types, virulence-associated gene profiles, and antimicrobial-resistance patterns of *P. multocida* for disease predilection.

Materials and methods

Sample collection

Altogether 1023 morbid tissue samples including lungs (676) and palatine tonsils (347) were collected at abattoirs in Harin ghata (West Bengal, 197), Ranchi (Jharkhand, 89), Kanpur and Bareilly (Uttar Pradesh, 342), Mumbai (Maharashtra, 232), and Guwahati (Assam, 163) from sacrificed pigs, aged < 6 months age (385) and an adults (638), during the period from January 2021 to April 2022. In the slaughterhouse, the vertical scalding of pig carcasses was done for 5 min using a double-walled tunnel, in which steam generated from a water bath in its bottom is blown over the pig carcasses through the ventilator. The 553 lungs showed varying degrees of pneumonic lesions, whereas 123 lungs were without appreciable gross lesions. A total of 347 palatine tonsils irrespective of gross lesions were collected for the detection of *P. multocida*. Above slaughterhouses were targeted because of the high turnover of pigs from densely pig-rearing areas.

Gross and histopathological investigation

Tissue samples (<0.5 cm thick) from the lungs and palatine tonsils, collected in 10% neutral buffered formalin (NBF) and over ice, were used for pathological and molecular biology investigations, respectively. The formalin-fixed tissues were processed for making paraffin-embedded tissue sections of 4–5 µm thickness with a semiautomatic microtome (Leica RM2245), followed by staining with routine haematoxylin and eosin stains (Luna 1968). The duplicate paraffin sections on poly-L-lysine-coated glass slides were processed for the immunostaining method. Tissue changes were photographed using an upright light microscope (Olympus, BX3, Japan and 5 Mega Pix Jenoptic Camera).

Immunohistochemical investigation

The formalin-fixed paraffin-embedded (FFPE) tissue sections of lungs and spleen were used for immunohistochemical (IHC) investigation. The antigen unmasking was achieved by immersing slides in 0.01 M citric buffer (pH 6.0) under microwave irradiation for 10 min, followed by quenching the endogenous peroxidases with 0.3% H₂O₂ solution in absolute methanol for 30 min. Then, the tissue sections were covered with 5% normal goat serum in phosphate-buffered saline (PBS) for the 30 min to block the background staining. The tissue sections were incubated overnight at 4°C with *in-house* raised rabbit polyclonal hyperimmune sera (raised against the whole cell antigen of *P. multocida*) at dilution of 1:100 in PBS containing 1% bovine serum albumin (BSA) in a humidified chamber followed by incubation with secondary antibody goat anti-rabbit IgG-HRP (Santacruz, Cat# sc-2005) diluted at 1:100 for 1 h. For the development of colour, the tissue sections were covered with 3,3'-Diaminobenzidine (DAB)-enhanced liquid substrate (Sigma, Cat# D3939) for 15 s followed by counterstaining with the Mayer's haematoxylin for the 30 s. Finally, the slides were mounted with CC mount™ (Sigma-Aldrich, Cat #C9368). Negative control tissue sections from each block of both disease and healthy pig tissues (lungs and palatine tonsil) were substituted with similar dilutions of PBS with 1% BSA, and with rabbit isotype control antibodies (Thermo Fischer Scientific, USA) to ruling out the false positive staining.

Bacterial isolation and biochemical characterization

The lungs and palatine tonsils were processed for the isolation of *P. multocida* by plating on 5% sheep blood agar plate and MacConkey agar plate followed by overnight incubation at 37°C. The nonhaemolytic, medium to big-sized dewdrop-like mucoid colonies on blood agar plate, no growth in MacConkey agar plate, and Gram-negative coccobacilli on Gram staining were suspected for *P. multocida*, which further subjected to a panel of biochemical tests-like catalase test, oxidase test, indole test, Methyl Red and Voges-Proskauer Test (MR-VP test), citrate utilization test, urease test, nitrate reduction test, sugar fermentation tests such as glucose, mannitol, sucrose, mannose, maltose, arabinose, lactose, dulcitol, salicin, inositol, and trehalose sugars.

Antibiotic sensitivity test

Antibiotic sensitivity testing (ABST) was performed on 57 lungs isolates of *P. multocida* by the modified disc diffusion method (Bauer 1966) using 12 antibiotics such as amoxicillin (30 µg), ampicillin (10 µg), enrofloxacin (5 µg), erythromycin (15 µg), chloramphenicol (30 µg), neomycin (30 µg), doxycycline (30 µg), colistin (10 µg), penicillin (10 µg), tetracycline (30 µg), levofloxacin (5 µg), and tobramycin (10 µg).

Molecular detection of *P. multocida*

Extraction of genomic DNA

DNA was isolated from the lungs and palatine tonsils using DNeasy Blood and Tissue Mini Kit (Qiagen, USA) as per the manufacturer's instructions. DNA was isolated from the bacterial colonies by the Snap-chill method (Tang et al. 2009).

Capsular typing and virulence gene profiling

Polymerase chain reaction (PCR) confirmed *P. multocida* isolates were subjected to molecular typing by *P. multocida*-specific PCR (PM-PCR) using published primers. Virulence typing of *P. multocida* was conducted for 11 virulence-associated genes such as *tox*A, *has*R, *exb*B, *ptf*A, *pfh*A, *hsf*-1, *nan*B, *sod*C, *plp*B, *plp*E, and *oma*87. The primers used for the capsular typing and virulence gene profiling of *P. multocida* are presented in Table 1.

Mouse pathogenicity testing of untypeable capsular isolates

A total of 102 healthy Swiss albino mice of 6–8 weeks of age were inoculated with 18 h enriched brain heart infusion (BHI) cultures of 34 untypeable capsular isolates in volume of 0.1 ml per each mouse via intraperitoneal route, and the control group ($n = 3$) received 0.1 ml of PBS. All the mice were observed for clinical signs of the disease, systematically necropsied to record the pathomorphology (gross and histopathology) in various internal organs. The heart blood, lungs, and spleen were processed for the reisolation of *P. multocida*. Animal care and the experimental procedures were performed as per the recommendations and prior approval of the Institute IVRI-Animal Ethics Committee of IVRI, Izatnagar (IVRI-IAEC), and the guidelines set forth by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), India [Approval No. F.26–1/2018–19/JD (R)].

Statistical analyses

The prevalence of Pasteurellosis between the age groups was analysed by paired sample *t*-test using Graphpad Prism, Version 8.0. A value of $P < .0001$ was considered statistically significant. The association of resistance/susceptibility of antibiotics among the capsular serotypes of *P. multocida* was analyzed by multinomial logistic regression analysis test using SPSS 29.0 (SPSS Inc., Chicago). A value of $P < .05$ was considered statistically significant.

Results and discussion

Out of 1023 samples examined, 278 (27.1%) were positive for *P. multocida* yielding an amplified product of 460 bp (Fig. 1a). Out of 278 positive cases, 104 (pneumonic lungs-97; healthy-7) and 174 cases were positive from lungs and palatine tonsils, respectively. On microbiological investigation, 78 isolates (7.62%) were recovered. Out of 78 isolates, 55 isolates (pneumonic lungs-51; healthy-4) and 23 isolates were recovered from lungs and palatine tonsils (diseased-9; healthy-14), respectively. The similar finding was recorded in earlier reports (Radanovic et al. 2016, Liu et al. 2017). The higher prevalence in the older pigs (> 6 months, 68.7%) as compared to the younger pigs (< 6 months) ($P < .0001$) might be due to increased exposure of pigs (mainly horizontal transmission) to infection with the advancement of age (Mores 2006). The higher detection of *P. multocida* from pneumonic lungs suggests their role in pneumonia. The higher detection in palatine tonsils (65.01%) without apparent gross lesions suggests the carrier status, which might be attributed to the microenvironment of the tonsillar crypt favouring the colonization and growth of *P. multocida* (Ackermann et al. 1994, Townsend et al. 2000).

The isolates were positive for oxidase, catalase, indole production, and nitrate reduction, whereas negative for urease production test, citrate, MR-VP test, and gelatin liquefaction tests. All the isolates successfully fermented glucose, arabinose, maltose, mannitol, mannose, and sucrose, but failed to ferment dulcitol, inositol, salicin, trehalose, and lactose sugars. The similar biochemical characteristics were reported in various reports (Radanovic et al. 2016, Liu et al. 2017).

The capsular serotyping of 78 bacterial isolates showed the predominance of untypeable ($n = 34$; 43.58%) followed by cap A ($n = 26$, 33.33%), cap D (17; 21.79%), and cap B (1; 1.28%) yielding 1044, 760, and 657 bp product, respectively (Fig. 1a–d). The 34 isolates failed to show amplification for these five serotypes (A, B, D, E, and F), and thus, were assigned as untypeable isolates (Supplementary 1). Similarly, in morbid tissue samples, the predominant serotype belonged to untypeable capsular type (145, 52.15%) followed by cap A (87, 31.29%), and cap D (46, 16.54%).

In our case, the higher detection of untypeable capsular types in both molecular and microbiological methods, is an interesting finding. The detection of untypeable strains are reported from almost all animal hosts (Arumugam et al. 2011, Khoo et al. 2017) though limited in numbers (Rajkohwa et al. 2012). It should also be noted that the distribution of the different serotypes may vary from region to region, as well as from time to time in the same region. Recent report in China showed the high prevalence of cap A, whereas an earlier report before the 1990s showed the predominance of cap D (Liu et al. 2017). Similarly, in our study, the higher detection of untypeable capsular types followed by types A and D was observed, whereas the earlier reports showed the predominance of cap A followed by cap D in Indian pigs (Kalorey et al. 2008, Varte et al. 2014).

The distribution of virulence genes in all the serotypes of *P. multocida* is presented in Figure (Supplementary 2) and Table (Supplementary 3). The virulence genes *exb*B, *nan*B, *sod*C, *plp*B, and *oma*87 (100%) were detected in all isolates, whereas *has*R, *ptf*A, *pfh*A, *hsf*-1, and *plp*E genes were present in 91.3%, 92.8%, 22.0%, 54.6%, and 45.8% cases, respectively. Similarly, in 278 *P. multocida* DNA, *has*R, *ptf*A, *pfh*A, *hsf*-1, and *plp*E genes were present in 92.4%, 94.9%, 31.4%, 72.9%, and 19.2% cases, respectively. None of the isolates were positive for *tox*A gene. The 34 untypeable isolates showed the presence of *exb*B, *nan*B, *sod*C, *plp*B, and *oma*87 virulence genes (100%), which suggests their association with pathogenicity. These observations are congruent with the earlier workers (Sarangi et al. 2015, Garcia-Alvarez et al. 2017). The other virulence genes such as *has*R, *ptf*A, *pfh*A, *hsf*-1, and *plp*E genes were present in 76.1%, 71.4%, 47.6%, 85.7%, and 23.8% cases, respectively. The low prevalence of *pfh*A gene is corroborated with the previous report in Indian pig (Rajkhwa 2015). On contrary, *pfh*A gene was present in all the pathogenic strain of capsular type A isolates from Brazilian pigs (Oliveira Filho et al. 2018). Similarly, a higher detection of *hsf*-1 gene (85.7%), an auto-transporter adhesin in a common avian clone, Pm70 (May et al. 2001), was noted in untypeable capsular isolates, which might indicate its important role in host–pathogen interaction (Sarangi et al. 2015, Kim et al. 2019).

The antibiotic assay of 78 isolates is presented in Table 2. The study revealed 100% sensitivity to chloram-

Table 1. Details of primers and citations used for the detection, capsular typing, and virulence-associated gene profiling in porcine isolates of *P. multocida*.

S. No.	Gene	Purpose	Primer	Primer sequence (5'-3')	Product size (bp)	Reference
1	Cps	Molecular diagnosis	cps F cps R	GCT GTA AAC GAA CTC GCC AC ATC CGC TAT TTA CCC AGT GG	460	Townsend et al. (2000)
2	cap A	Serotyping	cap A-F cap A-R	TGCCAAATCGCAGTCAG TTGCCATCAATTGTCA GTG	1044	Townsend et al. (2000)
3	cap B	Serotyping	cap B-F cap B-R	CATTATCCAAGCTCCACC GCCCCGAGAGTTTCAATCC	760	Townsend et al. (2000)
4	cap D	Serotyping	cap D-F cap D- R	TTACAAAAGAAAGACTAGGAGCCC CATCTACCCACTCAACCATATCAG	657	Townsend et al. (2000)
5	cap E	Serotyping	cap E-F cap E-R	TCCGCAGAAAATTATTGACTC GCTTGCTGCTTGATTTTGTG	511	Townsend et al. (2000)
6	cap F	Serotyping	cap F-F cap F-R	AATCGGAGAACGCAAAATCAG TTCCGCCGTCAATTACTCTG	851	Townsend et al. (2000)
7	tox A	Virulence gene profiling	tox A F Tox A R	CTTAGATGAGCGACAAGG GAATGCCACAGCTCTATAG	864	Aski and Tabatabaei (2016)
8	hasR	Virulence gene profiling	hasR F HasR R	CATATCGGATCCAGAGCGTAT GGCGCAITTGCAA TAGATCGAGCTCAGCAACAAA GCGCTCCATCT	430	Khamesipour et al. (2014)
9	exbB	Virulence gene profiling	exbB F exbB R	TTGGCTTGATTTGAACGC TGCAGGAATGGCGACTAA A	282	Aski and Tabatabaei (2016)
10	ptfA	Virulence gene profiling	ptfA F PtfA R	AGGATCCATGAAAAAGCCATT GGAGCTCTATGGCAAAATCCTG	435	Aski and Tabatabaei (2016)
11	pflA	Virulence gene profiling	pflA F PflA R	TAAGCCTATCGGTCAAGTGG GATAAATCTACCCCGTCTCT	805	Aski and Tabatabaei (2016)
12	hsf-1	Virulence gene profiling	hsf-1 F hsf-1 R	TTGAGTCGGCTGTAGAGTTCG ACTCTTTAGCAGTGGGACACCTC	628	Aski and Tabatabaei (2016)
13	nanB	Virulence gene profiling	nanB F nanB R	GTCTATAAGTGACGCCGA ACAGCAAGGAGACTGTCC	585	Aski and Tabatabaei (2016)
14	sodC	Virulence gene profiling	sodC F sodC R	AGTTAGTAGCGGGTTGGCA TGGTGTGGGTGATCATCATG	236	Aski and Tabatabaei (2016)
15	plpB	Virulence gene profiling	plpB F plpB R	TTTGTTGGTGGGTATGTCTCT AGTCACTTTAGATTGTGGTAG	298	Aski and Tabatabaei (2016)
16	plpE	Virulence gene profiling	plpE F plpE R	CCATGGCATGAAACAAATCGTTTAA AAGCTTTGTGCTTGACTTTTTTC	1010	Ewers et al. (2006)
17	oma87	Virulence gene profiling	oma87 F oma87 R	ATGAAAAAATTTTAATTGCGAGC TGACTTGCCGAGTTGCATAAC	949	Aski and Tabatabaei (2016)

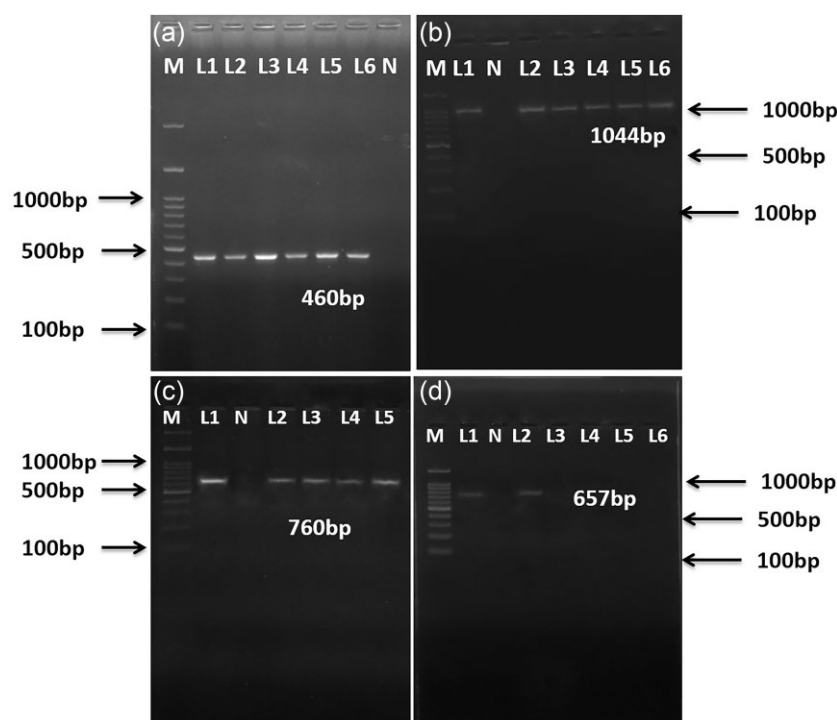


Figure 1. Molecular detection along with capsular typing of *P. multocida* isolates. (a) Lane- M: ladder (100 bp), L1: positive control, L2–L6: positive for *P. multocida*, and N: negative control. (b) Lane- M: ladder (100 bp), L1: positive control, N: negative control, and L2–L6: cap A. (c) Lane- M: ladder (100 bp), L1: positive control, N: negative control, and L2–L5: positive for cap D. (d) Lane- M: ladder (100 bp), L1: positive control, N: negative control, L2: positive for cap B, and L3–L6: negative for cap B.

Table 2. Antibiotic sensitivity/resistance pattern of various capsular serotypes of porcine isolates of *P. multocida*.

Capsular types	Antibiotic susceptibility	Positive strain percentage (%) to individual antibiotics											
		AMX	AM	E	EX	C	NEO	DOXY	CL	P	TET	LEVO	TOB
Untypeable (<i>n</i> = 34)	S	50.0	55.8	52.9	47.0	85.2	26.4	76.4	94.1	52.9	29.4	70.5	85.2
	I	14.7	26.4	41.1	29.0	14.7	61.7	23.5	5.8	32.3	44.1	23.5	14.7
	R	35.2	17.6	5.8	23.5	0.0	11.7	0.0	0.0	14.7	26.4	5.8	0.0
	S	42.3	53.8	46.1	61.5	69.2	50.0	69.2	92.3	46.1	53.8	84.6	80.7
cap type A (<i>n</i> = 26)	I	26.9	19.2	50.0	15.3	30.7	38.4	30.7	7.6	38.4	19.2	11.5	19.2
	R	30.7	26.9	3.8	23.0	0.0	11.5	0.0	0.0	15.3	26.9	6.2	0.0
	S	17.6	7.0	4.0	15.0	17.0	8.0	41.1	100.0	26.4	23.5	42.3	47.0
cap type D (<i>n</i> = 17)	I	5.8	5.0	11.0	2.0	0.0	9.0	11.5	0.0	15.3	26.9	17.6	6.2
	R	26.4	5.0	2.0	0.0	0.0	0.0	0.0	0.0	11.5	5.8	0.0	0.0
	S	100.0	0.0	0.0	0.0	1.0	0.0	100.0	100.0	100.0	100.0	0.0	100.0
cap type B (<i>n</i> = 1)	I	0.0	100.0	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	100.0	0.0
	R	0.0	0.0	0.0	1.0	0.0	100.0	0.0	0.0	100.0	0.0	0.0	0.0

AMX: amoxicillin, AM: ampicillin, E: erythromycin, EX: enrofloxacin, C: chloramphenicol, NEO: neomycin, DOXY: doxycycline, CL: colistin, P: penicillin, TET: tetracycline, LEVO: levofloxacin, and TOB: tobramycin.

phenicol, doxycycline, colistin, and tobramycin irrespective of their serotypes. Similar findings were also reported in earlier reports (Sellyei et al. 2009, Rajkohwa et al. 2012, Sarangi et al. 2015). The untypeable isolates showed highest level of resistance to amoxicillin (35.2%), followed by tetracycline (26.4%) and enrofloxacin (23.5%). The association of various antibiotics among the capsular types of *P. multocida*, which was analysed by multinomial logistic regression analysis, is presented in [Supplementary 4](#). The untypeable capsular types showed significant ($P < .05$) susceptibility towards ampicillin. The significant association ($P < .05$) of amoxicillin in terms of resistance and susceptibility was observed in untypeable cap-

sular types while only resistance was observed in case of type D isolates. The cap type A isolates showed significant resistance/susceptibility ($P < .05$) towards erythromycin. The untypeable capsular isolates showed significant ($P < .05$) susceptibility towards levofloxacin. The similar observations were documented in recent reports (Amaral et al. 2019, Vu-khac et al. 2020). The multidrug resistance might be attributable to the use of antibiotic additives in animal feed and the extensive use of antimicrobial agents in veterinary medicine. Moreover, the selective pressure by antimicrobial agents or due to transposon-mediated drug resistance (Kehrenberg et al. 2001), or association with certain genes might be

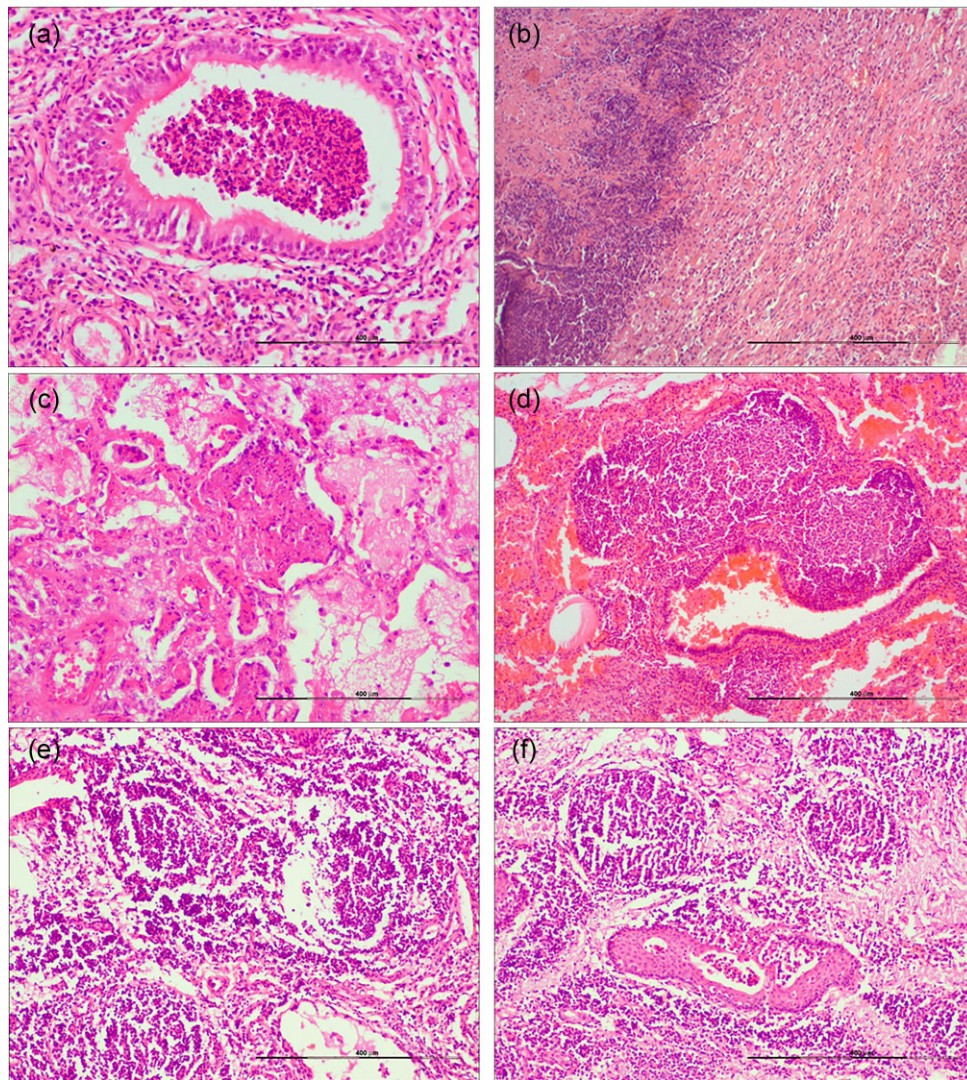


Figure 2. Microscopic lesions in pigs naturally infected with *P. multocida* showing: (a) Acute suppurative bronchopneumonia, lungs. (b) An area of abscess walled off by a thick wall with inflammatory cellular exudates, lungs. (c) Fibrinous pneumonia showing fibrinocellular exudates within the alveoli, lungs. (d) BALT hyperplasia suggestive of Enzootic pneumonia. (e) Depleted lymphoid follicles, palatine tonsil. (f) Necrosed tonsillar cryptic epithelium with luminal cellular exudates. H&E $\times 100$, scale bar = 400 μm .

responsible for mutation in 16S rRNA (Kehrenberg and Schwarz 2007).

Out of 553 pneumonic lungs, *P. multocida* detected in lung lesions of 97 PCR-positive cases, were further grouped into six categories according to the histopathological classification. The most frequent finding was acute bronchopneumonia (36), showing different types of exudation in alveoli and bronchioles, including neutrophils or a mixed cell population of mononuclear cells (Fig. 2a). *Pasteurella multocida* was detected in the lungs showing lesions of subacute bronchopneumonia and chronic bronchopneumonia in 5 and 28 cases, respectively. In the chronic suppurative bronchopneumonia cases, the lungs showed areas of abscessation surrounded by a thick wall of fibrino-cellular reaction with fibrosis of the alveolar septa (Fig. 2b). *Pasteurella multocida* DNA was detected in 13 cases of fibrinous pneumonic lungs showing infiltration of dead degenerated neutrophils admixed with fibrin in the bronchiolar lumen (Fig. 2c). Two cases of lungs showing necrotizing pneumonic lesions were positive of *P. multocida*. The lungs with interstitial pneumonia (acute-2; chronic

interstitial pneumonia-4; broncho-interstitial pneumonia-3) were positive for *P. multocida*. The lungs with Enzootic pneumonic lesions (four cases) showing congestion, haemorrhages, and bronchiolar-associated lymphoid tissue (BALT) hyperplasia, were positive for *P. multocida* (Fig. 2d). The lungs with the majority of untypeable capsular type isolates showed the predominant microscopic lesions of bronchopneumonia. The similar lesions were documented in earlier reports (Hansen et al. 2010, Paladino et al. 2017). The palatine tonsils showed lymphoid depletion in majority of the cases (Fig. 2e) with presence of necrotic cellular exudates in the cryptic lumen (Fig. 2f).

A total of 92 lungs and 127 palatine tonsils showed the distribution of bacterial antigens. The affected lungs revealed the abundant immunolabelling of antigen in alveolar macrophages (Fig. 3a), bronchial epithelium, and bronchial luminal neutrophils, and macrophages (Fig. 3b). The strong immunoreactivity in the macrophages could be due to its participation in mediating phagocytosis and cytokine production (Oliveira Filho et al. 2018). In the palatine tonsil, the cellular exudates (Fig. 3c), necrosed cryptic epithelium (Fig. 3d), and

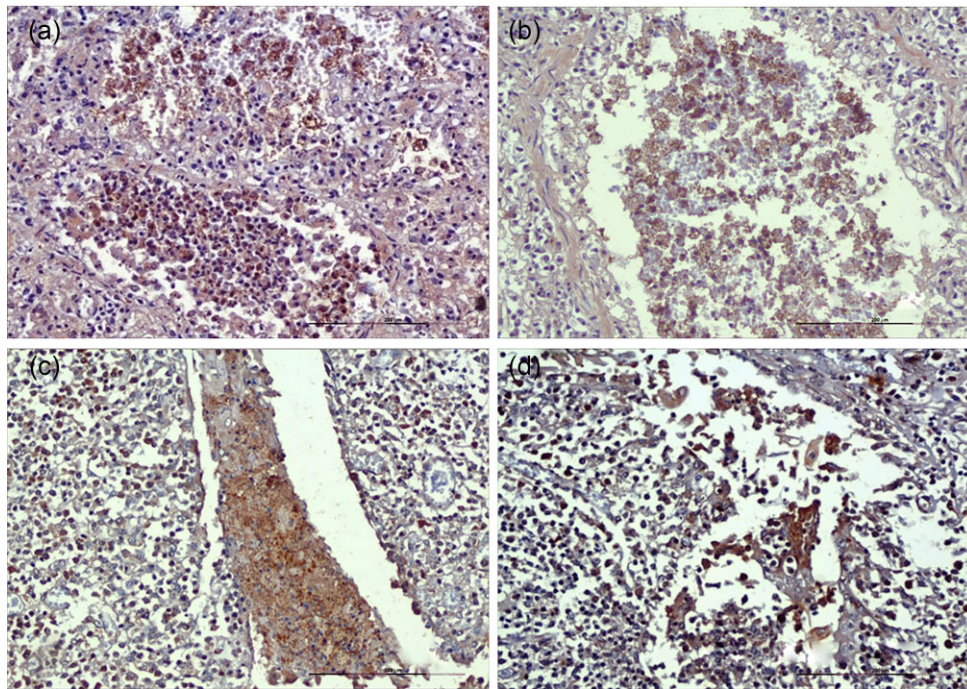


Figure 3. Immunolocalization of *P. multocida* antigen as indicated by dark brown colour signals in (a, b) Alveolar macrophages, lungs. (c) Cellular exudates, palatine tonsil. (d) Necrosed cryptic epithelium with lymphocytes, palatine tonsil. Immunoperoxidase staining (indirect method, ABC), hematoxylin counters. IHC, $\times 200$.

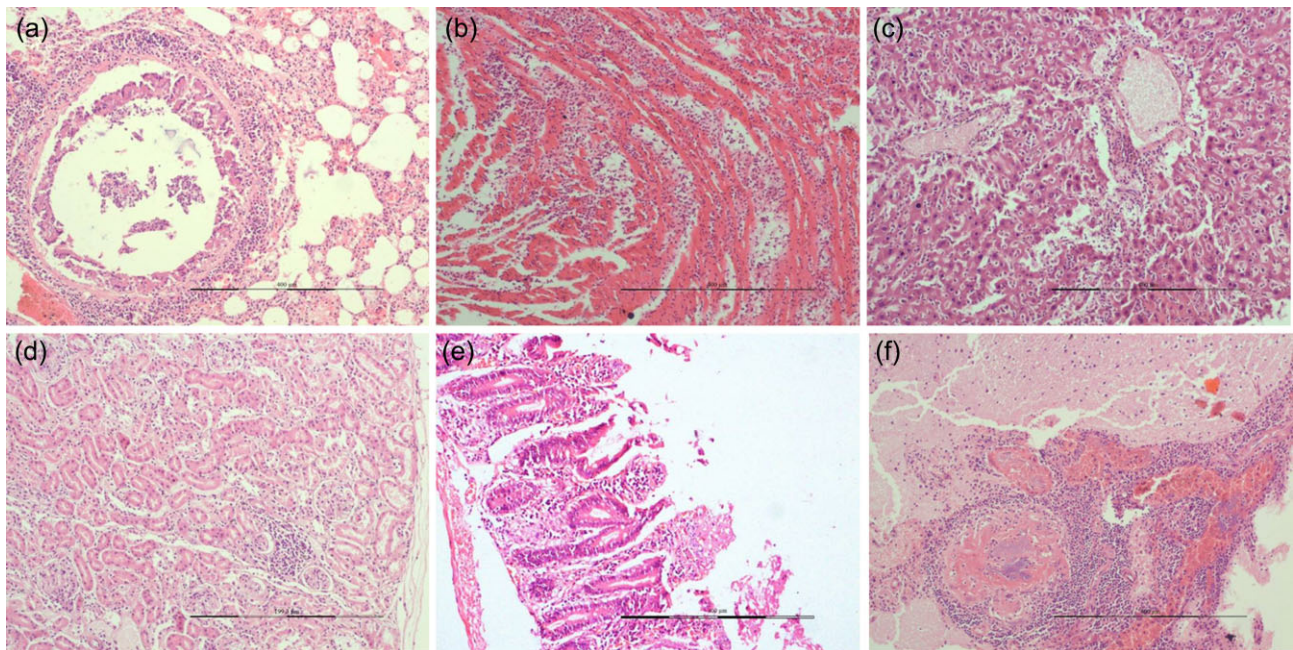


Figure 4. Histopathological alterations in mice inoculated with untypeable capsular serotype of *P. multocida* showing (a) Bronchopneumonia, lungs. (b) Suppurative myocarditis, heart. (c) Hepatitis, liver. (d) Nephritis, kidney. (e) Enteritis, intestine. (f) Leptomeningitis, brain. H&E $\times 100$.

the lymphocytes revealed the immunolabelling for bacterial antigens, which might be due to the subsequent lymphogenous route to the draining lymph nodes and via the hematogenous routes to the various organs.

The pathogenicity testing of untypeable capsular isolates showed the deaths of 38 mice within 24 h, and 23 within

48 h, while 8 mice died at 96 h. None of the isolates recovered from palatine tonsils were lethal to the mice. The histopathological lesions in mice inoculated with untypeable capsular serotypes are presented in Fig. 4. Microscopically, the mice showed the lesions of bronchopneumonia (Fig. 4a), suppurative myocarditis (Fig. 4b), hepatitis (Fig. 4c), nephritis

(Fig. 4d), enteritis (Fig. 4e), and leptomeningitis (Fig. 4f). On contrary, the control mice did not show gross and microscopic changes in the vital organs. The survived animals after sacrificed showed the mild vascular changes such as congestion and haemorrhages in lungs, heart, and kidney. The pure cultures of *P. multocida* were isolated from the heart blood, lungs, and spleen of inoculated mice. The similar observations has been documented in several studies (Khaleel et al. 2014, Sahoo et al. 2020).

Conclusion

The high detection of untypeable serotypes of *P. multocida* as compared to other capsular serotypes is an interesting finding. Further, the virulence gene profiling and mouse pathogenicity testing of these isolates confirm their pathogenicity. The findings of this study provide useful data to inform control strategies for respiratory disease; and are relevant to future work on the development of effective vaccines.

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Ethics approval

We have complied with the ARRIVE guidelines.

Supplementary data

Supplementary data is available at LAMBIO Journal online.

Conflict of interest: None declared.

Author contributions

Monalisa Sahoo (Conceptualization [lead], Investigation [equal], Project administration [lead], Visualization [lead], Writing – original draft [lead], Writing – review & editing [supporting]), Suraj Baloni (Formal analysis [equal], Investigation [supporting], Methodology [equal], Resources [lead]), Jigarji C. Thakor (Data curation [lead], Formal analysis [lead], Investigation [equal], Resources [equal]), Pradeep Kumar (Formal analysis [equal], Investigation [supporting], Methodology [equal]), Prasad Thomas (Investigation [supporting], Supervision [equal], Visualization [lead]), Viswas K. Nagaleekar (Supervision [equal], Validation [equal]), Kuldeep Dhama (Data curation [equal], Formal analysis [equal], Writing – review & editing [supporting]), Rajendra Singh (Conceptualization [supporting], Project administration [supporting], Writing – original draft [supporting], Writing – review & editing [supporting]), Karam P. Singh (Methodology [supporting], Supervision [supporting], Validation [supporting]), Saminathan Mani (Data curation [supporting], Supervision [supporting], Validation [equal]), Salaudinn Qureshi (Formal analysis [supporting], Resources [equal]), Ajay Kumar (Data curation [supporting], Formal analysis [supporting], Resources [equal]), Shailesh K. Patel (Formal analysis [supporting], Investigation [supporting], Resources [supporting]), Jitendra K. Biswal (Data curation [equal], Formal analysis [equal], Validation [equal]), and Nihar R. Sahoo (Data curation [supporting], Su-

pervision [supporting], Writing – review & editing [supporting]).

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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